

INTERNATIONAL STANDARD

**ISO
21043-1**

First edition
2018-08

Forensic sciences — Part 1: Terms and definitions

Criminalistique —

Partie 1: Termes et définitions



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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

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For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see the following URL: www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 272, *Forensic sciences*.

A list of all parts in the ISO 21043 series can be found on the ISO website.

Introduction

The development of international standards for forensic science is important to enhance the reliability, transparency and confidence in forensic evidence. Standards will harmonize work practices to facilitate forensic facilities from different countries to work collaboratively in response to cross border investigations. They will also enable facilities from different countries to support one another in the event of a catastrophic event that exhausts a country's capabilities.

Standards will also facilitate the exchange of forensic results, information and intelligence, including the sharing of databases, to ensure forensic services are fit for purpose. Standardization in the manner that forensic evidence is collected, analysed, interpreted and reported is critical to the common approach to how evidence is utilized. This enables the sharing of information and intelligence among jurisdictions, to successfully exonerate the innocent or prosecute perpetrators.

Consistent and accepted standards within the forensic community will benefit all users of the criminal justice system including members of the public as well as legal and forensic practitioners. Recent issues in various countries illustrate the benefits that standards could provide, particularly in the pattern matching sciences such as hair analysis, DNA interpretation and fingerprints.

Conformance to a platform of relevant standards for law enforcement and forensic disciplines would ensure that methodologies are robust, repeatable and validated, and that training across jurisdictions is consistent. This would have a direct bearing on the quality of scientific evidence presented in the courts, and would increase the likelihood of successful justice outcomes.

This document is part of a series which, when completed, will include the different components of the forensic process from scene to courtroom (as illustrated in [Figure 1](#)). The series describes primarily "what" is standardized, not the "how" or "who". Best practice manuals and standard operating procedures should describe "how" the requirements of this document would be met. National regulations and policies will determine "who" should meet the requirements of this series.

This document provides a vocabulary that standardizes the use and meaning of terms associated with the forensic sciences. These terms are drawn from standard terms used in the forensic science processes, as contained in the core forensic standards developed to date. Discipline-specific standards will include their own specific vocabulary.

This document is organized alphabetically and follows natural (English) word order wherever possible. The source documents for the definitions are International Standards developed by ISO/TC 272. Where possible, users are advised to verify whether a more recent edition of the source document has been published, which contains an updated version of the terms and definitions. Future versions of this document will include updated terms and definitions consistent with the source documents at the time the revision is prepared.

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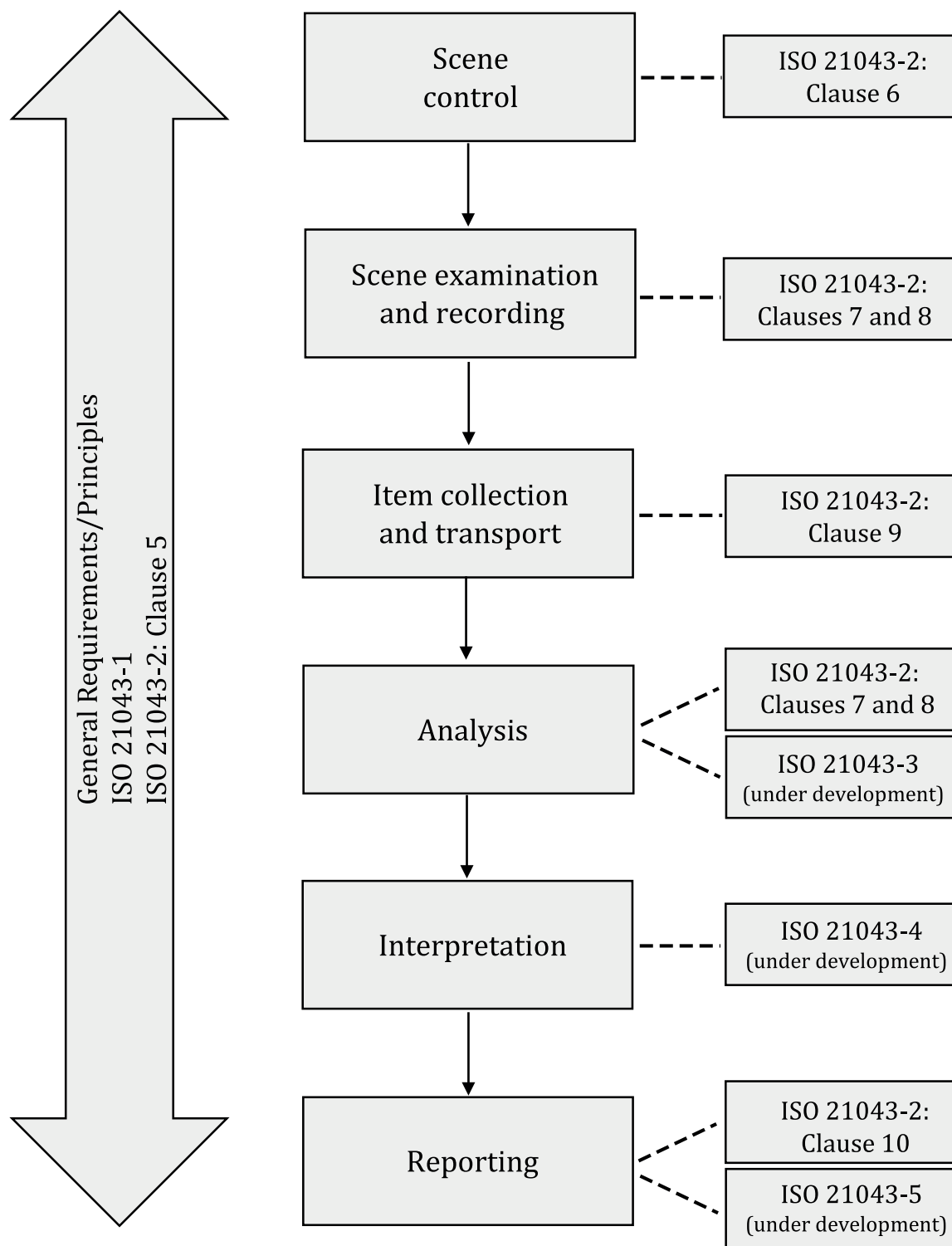


Figure 1 — Relationship between the various components in the forensic process and the clauses within the ISO 21043 series

Forensic sciences —

Part 1: Terms and definitions

1 Scope

This document defines terms used in the ISO 21043 series of standards.

2 Normative references

There are no normative references cited in this document.

3 Terms and definitions

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <http://www.electropedia.org/>

3.1

chain of custody

chronological *record* (3.23) of the handling and storage of an *item* (3.19) from its point of collection to its final return or disposal

Note 1 to entry: Chain of custody is one element that contributes to the integrity of an *item* (3.19).

3.2

consumable

single use or limited use material which is used in the *forensic process* (3.15)

3.3

contamination

undesirable introduction of a substance to an *item* (3.19) at any point in the *forensic process* (3.15)

Note 1 to entry: This includes undesirable transfer of a substance within an item or between *items* (3.19) (also referred to as cross-contamination).

3.4

control sample

material with known properties, analysed in order to evaluate the performance of the test and to ascertain that the data obtained are valid

3.5

customer

client, authority, organization or person(s) requesting the *forensic* (3.14) services

3.6

document

information and the medium on which it is contained

EXAMPLE Specification, procedure document, policy, instruction or form, drawing, *record* (3.23), *report* (3.26), standard, flowchart.

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Note 1 to entry: The medium can be paper, magnetic, electronic or optical computer disc, photograph, or a combination thereof.

Note 2 to entry: A set of documents, for example specifications and *records* (3.23), is frequently called “documentation”.

[SOURCE: ISO 9000:2015, modified]

3.7 document control

system to establish, approve, issue, maintain, distribute, use, change, store, secure and dispose of *documents* (3.6)

Note 1 to entry: *Records* (3.23) are covered under *record control* (3.24).

3.8 enhancement

process to reveal, intensify, magnify or clarify an observation or result

3.9 examination

act or process of observing, searching, detecting, recording, prioritizing, collecting, analysing, measuring, comparing and/or interpreting

Note 1 to entry: Examination can include collecting *items* (3.19) from persons.

3.10 examination strategy

plan developed to specify the requirements and activities for the *examination* (3.9) phase of a *forensic process* (3.15)

Note 1 to entry: An examination strategy can be applied to an investigation *scene* (3.29), the *examination* (3.9) of an *item(s)* (3.19) or sequential multi-discipline testing on an *item(s)* (3.19).

3.11 facility

physical environment used to protect the *item* (3.19) integrity, conduct testing, or support any other aspect of the *forensic process* (3.15)

EXAMPLE Buildings, designated area, tents, storage areas, mobile office or laboratories and vehicles.

3.12 finding

information concluded as a result of an *examination* (3.9)

3.13 first responder

first law enforcement personnel arriving at a *scene* (3.29) responsible for immediate action

3.14 forensic

related to methods, techniques and processes used to establish conclusions and/or opinions, facts and *findings* (3.12), which can be used for legal proceedings

3.15 forensic process

set of interrelated or interacting *forensic* (3.14) activities

Note 1 to entry: The forensic process can include *item* (3.19) handling and control, *examination* (3.9), reporting, and testimony.

3.16

forensic service provider

organization or individual that conducts and/or supplies *forensic* (3.14) services

3.17

impartiality

objectivity, neutrality and fairness along with the awareness of bias and reduction of prejudice

Note 1 to entry: It is important for impartiality that possible conflicts of interest are declared, resolved or do not exist.

3.18

interpretation

use of professional judgement to provide conclusions and/or opinions on hypotheses, based on *findings* (3.12) and information gathered through the *forensic process* (3.15)

3.19

item

object, substance or material that is collected, derived or sampled as part of the *forensic process* (3.15)

3.20

peer review

evaluation of the *reports* (3.26), *examinations* (3.9), notes, data and *findings* (3.12) by others competent in the same field to assess that there is an appropriate and sufficient basis for the conclusions and/or opinions

3.21

quality

degree to which a set of inherent characteristics fulfils a requirement

[SOURCE: ISO 9000:2015, 3.6.2, modified]

3.22

recognition

identification of *items* (3.19) or characteristics that can have potential *forensic* (3.14) value

3.23

record

document (3.6) providing information on observations or activities performed

3.24

record control

system used to ensure traceability and to identify, store, protect, retrieve, retain and dispose of *record(s)* (3.23)

3.25

reference sample

reference specimen

material from a known source used for comparison purposes in a *forensic process* (3.15)

Note 1 to entry: This includes biological material taken from a person (e.g. hair, blood, saliva), and non-biological material (e.g. carpet, paint, fabric).

Note 2 to entry: A reference sample can be taken from an *item* (3.19) in order to ensure the *substrate* (3.31) has no adverse effect on the performance of the test.

Note 3 to entry: A reference sample can be used to address the question of the source of an unknown *sample* (3.28), one would compare the unknown *sample* (3.28) with a reference sample.

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3.26

report

communication of outcomes of the *forensic process* (3.15)

EXAMPLE Observations, *findings* (3.12), *interpretations* (3.18), conclusions and/or opinions.

3.27

risk assessment

systematic process of the identification and evaluation of perceived or potential risks throughout the *forensic process* (3.15)

3.28

sample

portion drawn from a whole or population for the purpose of *examination* (3.9)/testing, not necessarily representative of the whole

Note 1 to entry: This includes biological material taken from a person (e.g. hair, blood, saliva).

EXAMPLE A representative sample from a powder for drug testing.

3.29

scene

place or object that is subject to and/or requires forensic *examination* (3.9)

Note 1 to entry: A crime scene is a common description of a scene where a presumed crime has been committed.

Note 2 to entry: The scene can be a person or an animal.

3.30

standard operating procedure

SOP

authorized, documented specified way to carry out an activity or process

[SOURCE: ISO 9000:2015, 3.4.5, modified]

3.31

substrate

surface or material upon which a substance is deposited

Bibliography

- [1] ISO 9000:2015, *Quality management systems — Fundamentals and vocabulary*

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ICS 07.140

Price code A

INTERNATIONAL STANDARD

**ISO
21043-2**

First edition
2018-08

Forensic sciences —

Part 2: Recognition, recording, collecting, transport and storage of items

Criminalistique —

*Partie 2: Reconnaissance, enregistrement, collecte, transport
et stockage des traces et autres éléments d'intérêt*



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Foreword

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Introduction

This document is part of a series which, when completed, will include the different components of the forensic process from scene to courtroom. This document establishes requirements designed to safeguard the reliability of results produced by the forensic process. It includes measures to protect items during the process of recognition, recording, collecting, transport and storage, which apply throughout the forensic process, both in the field and in the facility. The requirements are designed to protect items from loss, degradation, contamination and to minimize the negative impact of handling or packaging on downstream process.

[Annexes A, B](#) and [C](#) are for information only.

The notes given provide clarification of the text, examples and guidance. They do not contain requirements and do not form an integral part of this document.

Forensic sciences —

Part 2: Recognition, recording, collecting, transport and storage of items

1 Scope

This document specifies requirements for the forensic process focusing on recognition, recording, collection, transport and storage of items of potential forensic value. It includes requirements for the assessment and examination of scenes but is also applicable to activities that occur within the facility. This document also includes quality requirements.

This document is not applicable to procedures for the recovery of data from digital storage media which is covered by ISO/IEC 27037. However, the storage medium itself can yield additional items of forensic value (e.g. fingerprints or DNA).

[Annex D](#) shows the applicability of this document to the forensic process.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 21043-1, *Forensic sciences — Terms and definitions*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 21043-1 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <http://www.electropedia.org/>

4 Abbreviations

DNA	Deoxyribonucleic acid
PPE	Personal Protective Equipment

5 General requirements

5.1 Quality

Policies and/or procedures shall be documented and include:

- a) document control;

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- b) education and training requirements for personnel to ensure competence (see also [5.5](#));
- c) how corrective action is undertaken, recorded and reviewed;
- d) monitoring and recording conformance with policies and procedures;
- e) quality checks including peer review;
- f) record control;
- g) responsibility, authority and role of personnel;
- h) standard operating procedures;
- i) use of external service providers.

Policies and procedures within the scope of this document shall be readily accessible to relevant personnel, who shall be familiar with and operate within these requirements.

5.2 Requests for service

Customer requests shall be recorded and considered when developing the forensic examination strategy.

NOTE The customer requests and the scope of assignments are not always clear before the forensic examination has been started.

The customer should be informed and/or consulted on significant changes to the requested examination.

5.3 Impartiality and confidentiality

Policies regarding impartiality and confidentiality within the organization, its personnel and all activities shall be documented.

Information obtained should not influence the impartiality of the forensic process. Any information obtained that potentially impacts the impartiality of the forensic process shall be recorded.

5.4 Health and safety

Health and safety risk assessments shall be undertaken throughout the entire forensic process with appropriate actions to mitigate or control relevant risks. Identified relevant risks should be recorded and taken into consideration throughout the forensic process, especially during the development of the examination strategy.

NOTE Further information regarding potential risks and handling are respectively included in [Annex A](#) and [Subclauses 9.3](#) and [9.4](#).

5.5 Personnel

Roles and responsibilities of personnel shall be defined. Personnel involved in the forensic process, including external providers engaged by the forensic service provider, shall be competent to perform their role.

Requirements for competence of forensic service provider personnel shall include:

- a) an understanding of the different forensic disciplines;
- b) item handling, control, and chain of custody;
- c) loss, degradation, contamination or alteration prevention (see [Annex B](#) for additional information);
- d) recognition of items of potential forensic value;

e) specific examination and/or collection methods appropriate for the forensic process.

The minimum education, training and experience, and demonstrated competence shall be defined and documented.

Documented policies shall include maintenance of skills and expertise, such as continued education, ongoing training and reassessment of competence.

Competence of personnel shall be demonstrated and recorded.

5.6 Equipment and consumables

Equipment and consumables shall be suitable for intended use.

Forensic service providers shall identify equipment and consumables that can significantly impact the reliability of the result and have a documented procedure that should include:

- a) calibration and/or performance checks;
- b) cleaning;
- c) contamination prevention;
- d) maintenance;
- e) operational usage;
- f) traceability;
- g) working environment and storage requirements.

Maintenance, calibration and performance checks of such identified equipment shall be recorded.

5.7 Detection, screening and enhancement techniques

Documented procedures shall be in place for the use of detection, screening and enhancement techniques including appropriate quality control measures.

Equipment and reagents used for detection, screening and enhancement purposes that can significantly impact the reliability of the result shall be checked for performance on a scheduled basis, with the exception of single use kits. Performance checks of single use reagents should be undertaken for each batch where possible. Results of performance checks shall be recorded and evaluated against defined performance criteria.

5.8 Facilities

Facilities shall be suitable for their intended use.

Forensic service providers shall have documented procedures to reduce the risk of contamination. This shall include maintaining a clean environment only accessible by authorized personnel.

NOTE Further information regarding contamination minimization is included in [Annex B](#).

Forensic service providers shall have secure areas for the storage and handling of equipment and items.

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6 Initial response at scenes

6.1 First response

The forensic service provider shall provide guidance to first responders to minimize loss, degradation, contamination or alteration to secure and preserve the integrity of the scene.

Guidance should be accessible to non-law enforcement professionals who can be called to attend a scene.

NOTE Further information in relation to actions of first responders is included in [Annex C](#).

6.2 Forensic response

Relevant details of the incident should be obtained at the earliest opportunity to determine the nature of the forensic response required including the customer request(s), required personnel and equipment. This information should include the following:

- a) the nature of the incident;
- b) the location, time and date of the incident;
- c) the location of any secondary scenes.

Upon attendance at the scene, the following shall be reviewed:

- 1) the need for cordons and whether any existing cordons are adequate;
- 2) any specific safety hazards associated with the scene or items and the level of the PPE to be used during the examinations (the protective equipment should prevent contamination of the scene as well as protect the examiner);
- 3) information regarding who has had contact or association with the scene and for what purpose including whether anything has been moved, altered or touched;
- 4) information from persons involved or witnesses potentially relevant to the incident.

A record of persons who have entered a scene should be maintained. This can be maintained by law enforcement or forensic service provider personnel.

7 Forensic examination

7.1 General

All forensic examinations shall be systematic, impartial, appropriately planned and documented to optimize recognition and collection of items of potential forensic value. Alternative hypotheses shall be considered.

A coordinated approach should be used for examinations that require a multidisciplinary and/or complex response.

The examination process should aim to establish the facts about the course of events regarding the incident. This process should aim to provide information including:

- a) what happened;
- b) where it happened;
- c) when it happened;
- d) how it happened;

- e) who was/were involved.

7.2 Assessing a scene

The following factors should be considered when assessing a scene:

- a) mechanisms for securing the scene including entry/exit routes;
- b) areas within the scene that require examination;
- c) environmental conditions that would require prioritizing the processing of certain areas or items;
- d) examinations that need to be conducted before an item is handled and/or moved;
- e) whether extra coordination and/or specialist advice, equipment or assistance is required.

7.3 Examination strategy

A strategy shall be developed to conduct the forensic examination. Such a strategy shall depend on the case circumstances, professional judgement and information at hand and shall include the following:

- a) resources to be utilized;
- b) health and safety risks and appropriate precautions including PPE (see [Annex A](#) for additional information);
- c) protective measures required to prevent contamination including PPE (see [Annex B](#) for additional information);
- d) appropriate search method(s);
- e) items to be collected and requirements for collection;
- f) sequence of examinations;
- g) recording techniques to be utilized.

The examination strategy shall be reassessed as a result of new findings, insights, further information or change in circumstances.

7.4 Review and final closure of a scene

Prior to final closure of the scene, the examination strategy should be reviewed to determine if it has been completed and if the examination purpose has been fulfilled.

Consideration should be given to the potential for further examinations being required if circumstances change or further information comes to light. Security and integrity of the scene should be maintained if this is likely to occur.

The scene should be cleared to ensure that no collected items or equipment are left behind.

8 Recording

Items shall be recorded in their situational context prior to any disturbance. The recording should continue throughout the examination.

Records can include notes, diagrams, photographs or other electronic recordings. The records shall be of sufficient detail to allow the examiner or another appropriately trained examiner to report accurately:

- a) scene location, description and environmental conditions when relevant;

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- b) location of items, findings and observations of forensic interest;
- c) examinations that were undertaken, where and when they were undertaken and by whom;
- d) results of the examinations, both positive and negative;
- e) description of collected items, and unique identifiers associated with those items.

The forensic service provider shall have documented procedures to ensure all records that are subject to analysis or interpretation are suitable for the intended purpose.

EXAMPLE Procedures for photography of impressions.

9 Item handling and control

9.1 General

Documented policies shall be in place for the recognition, collection and subsequent management of items to ensure that:

- a) the chain of custody is maintained and recorded;
- b) the potential for analysis is optimized;
- c) the risk of loss, degradation, contamination or alteration is minimized (see [Annex B](#) for additional information);
- d) the security and integrity of the item is not compromised.

If a person within the forensic process suspects that an item might have been compromised, those circumstances shall be recorded.

If destructive handling or degradation of items is expected, then a detailed description of the item in its initial state shall be recorded.

9.2 Collecting

Documented procedures shall be in place for the collection of various types of items that are routinely collected.

Procedures should include:

- a) how the item is to be collected, derived or sampled;
- b) instructions for collection of reference samples, where necessary;
- c) measures to minimize the risk of loss, degradation, contamination or alteration;
- d) the quantity required to enable replicate testing where practical.

NOTE Further information regarding contamination minimization is included in [Annex B](#).

9.3 Packaging

Documented procedures shall be in place for the packaging of routinely collected items to minimize risks of loss, degradation, contamination or alteration.

Procedures should include:

- a) measures required to make items safe and secure, and to preserve them prior to packaging;

- b) the need for separate packaging of items where the packaging of items together is likely to compromise them;
- c) type of packaging to be used;
- d) the manner in which the packaging is sealed and secured;
- e) any measure that is required to preserve integrity, including during transport;
- f) provisions for short and long term storage.

Hazardous items collected shall be placed in an appropriate container.

When no procedure exists for non-routinely collected items, the principles in a) to f) should be applied.

9.4 Labelling

Documented procedures shall be in place for labelling all items. The packaging and/or item shall be marked with a unique identifier that allows the chain of custody to be tracked. The unique identifier shall enable the following details to be traced:

- a) a description of the item;
- b) the location from which the item was collected;
- c) the date the item was collected;
- d) the name or identifier of the person who collected the item;
- e) the case or investigation to which the item relates;
- f) any other relevant information that can be of interest to downstream processes, such as time of collection.

Hazardous items collected shall be marked with an appropriate warning label.

Labelling may also include any specific storage or handling requirements.

9.5 Item transport and storage

Documented procedures shall be in place to describe how items collected should be registered, transported and stored to maintain item integrity and the chain of custody. The item shall be traceable throughout the entire forensic process.

Transport and storage conditions should minimize loss, degradation, contamination or alteration due to environmental conditions such as temperatures, radiation and moisture.

Storage conditions should be considered upon reception of any item with a view to possible long-term storage and potential for later examinations. If the item is prone to degradation, appropriate measures, for example sampling, should be undertaken prior to long-term storage.

EXAMPLE The item of interest can be sampled and the substrate discarded.

Documented policies shall include how to manage items that are received in compromised or inappropriate packaging.

10 Reporting

The report shall be accurate, clear, unambiguous, impartial and suitable for intended use. The report should be provided to the customer.

The report should be approved before delivery to the customer and peer reviewed when necessary.

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Documented procedures shall be in place to ensure that the integrity of the written report is maintained, including updates, and that they are retained in a systematic, traceable and retrievable way.

Alterations to a report after delivery to the customer shall be recorded, and sent to the customer where applicable.

Annex A **(informative)**

Health and safety considerations

Scenes can occur in any type of environment. As a result, the risks cannot be readily apparent and can be uncontrolled. Prior to entering a scene, first responding forensic personnel should undertake a scan of the work area and identify potential risks. While not exhaustive, the following hazards can be encountered at scenes:

- a) biological;
- b) chemical and drugs;
- c) confined spaces;
- d) electrical wiring (unsafe wiring practices, dangerously overloaded circuits);
- e) exposure to UV light (Use of light sources);
- f) extreme temperatures;
- g) fire;
- h) inhalable dusts;
- i) living persons and animals associated with the scene (suspects, pets, livestock);
- j) mantraps (live wiring, explosives, tripwires connected to loaded firearms);
- k) physical (firearms, other weapons, sharps);
- l) psychological (exposure to traumatic circumstances or images);
- m) radiological;
- n) structural (unsafe floors, walls, heights and overheads);
- o) trip, slip or fall hazards;
- p) unexploded devices.

Risks should be assessed and precautions should be appropriate to the risks involved. The guiding principle is the protection of oneself and others.

Unidentified materials should be handled in such a manner as to avoid ingestion, inhalation and dermal exposure.

The level of PPE and the circumstances requiring PPE to be changed should be determined to mitigate the following risks:

- a) exposure of personnel to hazardous material and environments;
- b) exposure of other persons to hazardous material the personnel might have been exposed to;
- c) contamination of items.

PPE should be changed when necessary to minimize contamination.

Annex B **(informative)**

Contamination minimization

Care should be taken to mitigate contamination between involved persons, items, facilities, equipment, forensic personnel and/or scenes. Where there is a risk for contamination:

- a) items from victims, suspects, and scenes are collected and packaged separately;
- b) when possible, items from victim and suspect should be collected by different individuals. If not possible, the individual collecting the items should change PPE between collection events;
- c) collection equipment should be disposable and discarded after each use. If this is not possible, then the collection equipment is decontaminated after each use with an effective cleaning procedure;
- d) collection equipment and packaging should be free of detectable contamination regarding the type of material to be collected and type of analysis to be conducted;
- e) appropriate PPE is worn and changed when necessary to minimize contamination.

Annex C **(informative)**

First responder actions

Preservation of life and safety is paramount, including that of the first responder personnel. Then, as far as possible, the most important aspects of securing the scene are to preserve the scene with minimal contamination and disturbance of its original state. The first responder should, as far as practicable, proceed as follows:

- a) assess any possible health and safety hazards;
- b) identify any injured person(s) and provide assistance;
- c) assess possible entry and exit points;
- d) preserve the scene from loss, degradation, contamination or alteration by:
 - 1) clearing the scene of unauthorized persons,
 - 2) securing the scene (e.g. cordoning off) and preventing unauthorized entry,
 - 3) minimizing any contact with items of potential forensic value;
- e) start a scene log to record entry and exit of all personnel;
- f) make and record initial observations of the scene location and condition;
- g) secure items that would be lost prior to forensic personnel attendance;
- h) take appropriate notes of anything of possible interest to the investigators;
- i) preserve records, including any photographs or video footage;
- j) brief investigators and/or forensic service providers.

Annex D (informative)

Applicability of ISO 21043-2 to the forensic process

Figure D.1 is a general scheme of how this document applies to the forensic process. This process map is designed to aid in the depiction that the forensic process is not always linear and may require a looping back among some steps.

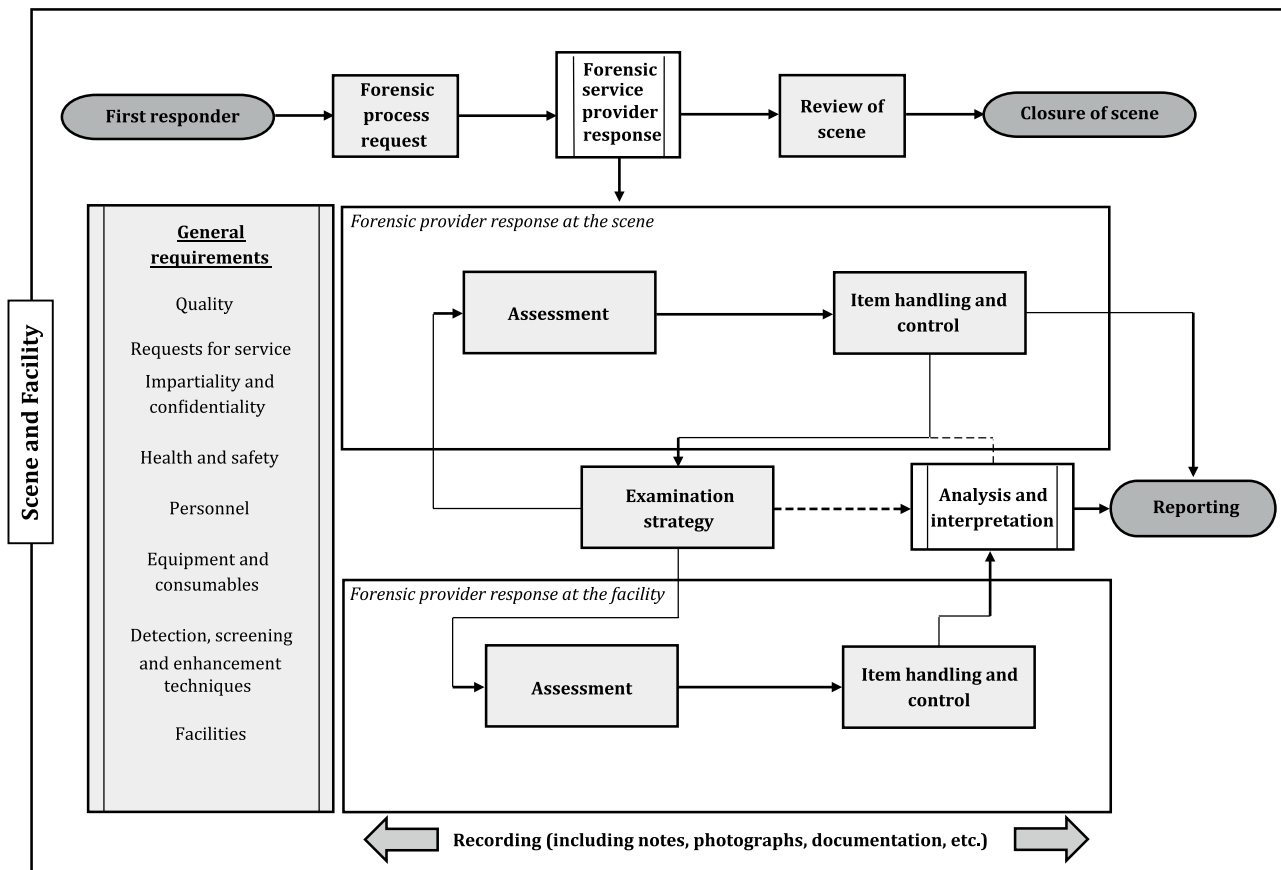


Figure D.1 — Map indicating how ISO 21043-2 applies to the forensic process

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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

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For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT), see the following URL: www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 272, *Forensic sciences*.

The following is a list of all parts in the ISO 21043 series:

ISO 21043-1, *Forensic Sciences – Part 1: Terms and definitions*

ISO 21043- 2, *Forensic Sciences – Part 2: Recognition, recording, collection, transport and storage of items*

ISO 21043- 3, *Forensic Sciences – Part 3: Analysis*

ISO 21043- 4, *Forensic Sciences – Part 4: Interpretation*

ISO 21043- 5, *Forensic Sciences – Part 5: Reporting*

Introduction

This document is part of a series of five that includes the different components of the forensic process from scene to courtroom. This document establishes requirements designed to safeguard the process for the analysis of items. The requirements are designed to ensure the use of suitable methods, proper controls, qualified personnel, and appropriate analytical strategies throughout the forensic analysis of items.

In this document, the following verbal forms are used:

- “shall” indicates a requirement;
- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or a capability.

Further details can be found in the ISO/IEC Directives, Part 2.

[Annexes A](#) and [B](#) are for information only.

The notes given provide clarification of the text, examples and guidance. They do not contain requirements.

Forensic Sciences —

Part 3: Analysis

1 Scope

This document specifies requirements and provides recommendations for the analysis of items of potential forensic value. It includes requirements and recommendations for the selection and application of suitable method(s) for the analysis to meet the needs of the customer and fulfil the request.

It is applicable to activities conducted by a forensic service provider that occur at the scene and within a facility.

The requirements facilitate the comprehensive, accurate, and reliable analysis of items.

This document is applicable to all disciplines of forensic science; however, it is not applicable to the recovery of digital data which is covered by ISO/IEC 27037^[1].

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 21043-2, *Forensic sciences — Part 2: Recognition, recording, collecting, transport and storage of items*

Numerous requirements listed in this standard are also described in ISO/IEC 17025:2017, *General requirements for the competence of testing and calibration laboratories*.

However, references to ISO/IEC 17025:2017 are for information purposes only [2].

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 21043-1, *Forensic Sciences – Part 1: Terms and definitions* apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>

4 General requirements

4.1 General

The procedures, including methods, for the analysis of items shall be documented, suitable for their intended use and, when applicable, include information regarding:

- a) measures that mitigate - the risk of loss, degradation, contamination and/or alteration of items;

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- b) sampling/selection of items;
- c) item preparation;
- d) selection of methods suitable to address the customer's request;
- e) use of reference and control samples;
- f) metrological traceability;
- g) assessment of measurement uncertainty;
- h) activities to demonstrate reliability of the observations;
- i) maintenance, operation, and calibration of equipment (hardware and software), including relevant performance parameters that can substantially impact the observations and their interpretation;
- j) environmental conditions that can substantially impact the observations and their interpretation;
- k) method limitations;
- l) use of any external service providers.

The examiner shall record the relevant observations, data, information and methods used during analysis. Information includes reference to uniquely identified equipment and consumables that can substantially impact the reliability of the observations.

Deviations from a procedure shall be recorded and the record retained.

Refer to ISO 21043-2 for further requirements for item handling and control, including collecting, packaging, labelling, transport, storage, and chain of custody.

4.2 Validation and verification of methods

Methods, including software, calculations and measurements, shall be validated prior to implementation. Methods previously validated elsewhere shall be verified at the facility where they are to be used, prior to implementation. Off-the-shelf analysis software used within its intended purpose and scope may be considered sufficiently validated, but shall be verified prior to implementation.

Known source items, comparable to those encountered in casework, shall be used for method validation and verification.

When changes are made to a validated method, the influence of such changes shall be determined and where they are found to affect the original validation, a new method validation shall be performed prior to implementation^[2].

Method validation should include the following performance characteristics, when applicable:

- a) precision;
- b) accuracy;
- c) limit of detection;
- d) limit of quantitation;
- e) specificity/selectivity;
- f) calibration;
- g) robustness;
- h) carry-over/contamination.

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Records of method validation and verification shall be retained, e.g., in the form of a validation plan and a validation report. These shall include:

- i) procedure used;
- j) specification of the requirements;
- k) determination of the performance characteristics of the method;
- l) observations obtained;
- m) statement on the validity of the method, detailing its suitability for the intended use, relevant limitations of the method and caveats about its use;
- n) person who performed the validation or verification.

NOTE See ISO/IEC 17025:2017, 7.2 *Selection, verification and validation of methods* for guidance.

4.3 Equipment and consumables

All equipment that can substantially impact the results shall be uniquely identified and records on the version of the software and firmware shall be retained, if applicable. Equipment shall be maintained and operated as per manufacturer's instruction or the forensic service provider's documented procedure.

Consumables that can substantially impact the observations shall be checked against pre-established acceptance criteria for proper/intended function before use. Where consumables form part of a batch supply, then batch testing where appropriate may be used. The checks shall be recorded and the record retained.

NOTE The frequency of the check can depend on the type of consumable and frequency of use.

Refer to ISO 21043-2 for further requirements for equipment and consumables.

4.4 Calibration and performance checks

Equipment and software that can substantially impact the observations shall be checked against pre-established acceptance criteria for proper/intended function prior to use and shall be periodically checked to support its continued use.

Records of calibration and/or performance checks shall be retained. These records shall include:

- a) the unique identity of the equipment;
- b) the version of the software and firmware used, if applicable;
- c) the identity of any reference or control samples used;
- d) the date of the calibration and/or performance checks and the identity of the personnel performing them;
- e) the results of the calibration and/or performance checks and the corresponding evaluation;
- f) actions taken when calibration and/or performance checks are outside the expected outcome.

4.5 Metrological traceability

When necessary for the reliability of observations, metrological traceability of measurements shall be established through appropriate calibration of measuring equipment and/or use of certified reference materials.

NOTE See ISO/IEC 17025:2017, 6.5 *Metrological Traceability* for guidance.

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4.6 Reference and control samples, collections and databases

The forensic service provider shall, when applicable, utilize reference samples, control samples, collections or databases in the analysis of items.

Reference and control samples shall be:

- a) uniquely identified; and
- b) traceable to certified reference materials or obtained from known sources.

The origin of samples or the relevant known characteristics of samples in collections and databases shall be documented.

Reference samples, control samples and collections shall be stored in a manner that mitigates the risk of loss, degradation, contamination or alteration.

Databases shall be stored in a manner that mitigates the risk of loss and unauthorized alteration.

Reference or control samples shall not be used beyond their expiry date, unless it has been checked that they remain fit for purpose beyond this date. The results of the check shall be retained.

4.7 Significant figures and measurement uncertainty

The forensic service provider shall have a documented procedure to estimate the measurement uncertainty of reported measurement observations. Estimation of measurement uncertainty shall be included in the validation of quantitative methods.

When estimating measurement uncertainty, all components of sampling and analysis that can substantially impact measurement uncertainty shall be taken into account. The major sources of uncertainty relating to a method shall be documented along with the overall uncertainty and an explanation of how the overall uncertainty was derived.

The forensic service provider shall have a documented procedure addressing the use of significant figures and the rounding of numerical values.

4.8 Performance monitoring

The forensic service provider shall have a documented procedure that addresses the monitoring of its performance which may include:

- a) proficiency testing;
- b) interlaboratory comparison;
- c) intralaboratory comparison.

Performance monitoring can be declared or blind.

The forensic service provider shall participate regularly in a relevant proficiency testing program or interlaboratory comparison, if available. Performance monitoring programs should include sets of examples typical of casework complexity. If an appropriate proficiency test or interlaboratory comparison does not exist, the forensic service provider should implement an intralaboratory comparison.

The forensic service provider should choose proficiency testing providers that meet the requirements of ISO/IEC 17043. The frequency of participation shall be based on risk and shall be sufficient to demonstrate continued performance and competence.

When participating in performance monitoring, the forensic service provider shall use the same procedures as it does for casework. The criteria specifying what constitutes sufficient performance shall be defined prior to the monitoring.

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The results of performance monitoring shall be compared to the expected results or pre-established criteria and, when there are significant deviations, action(s) shall be taken to address the deviations.

Performance monitoring records shall include:

- d) identifiers of personnel involved and how they were involved;
- e) the identity of the equipment used;
- f) an identifier for the performance monitoring test undertaken;
- g) the analyses undertaken and the observations obtained;
- h) an evaluation against expected observations or pre-established criteria;
- i) any corrective action(s) undertaken;
- j) the evaluation of any impact on previous and current analysis where necessary.

Performance monitoring should also include analysis of trends if multiple performance tests are undertaken.

5 Personnel

Roles and responsibilities of personnel shall be defined and documented. Personnel involved in the forensic process, including external providers engaged by the forensic service provider, shall be competent to perform their role.

Refer to ISO 21043-2 for further requirements for personnel.

6 Facilities

Facilities shall be suitable for their intended use. Facilities shall be accessible to authorized personnel only.

The forensic service provider shall have a documented procedure to mitigate the risk of contamination. Procedures shall include maintaining a clean environment to the extent necessary for the analysis.

Incompatible activities shall be separated (e.g., suitable separation of pre- and post-PCR activities within a DNA facility).

Refer to ISO 21043-2 for further requirements for facilities.

7 Environmental conditions

If environmental conditions (e.g., temperature, humidity, magnetic field) can substantially impact the reliability of the observations, they shall be monitored, recorded, and, if possible, controlled.

The forensic service provider shall have documented procedure(s) that address how to manage relevant environmental conditions including what actions to take if any of these go out of specification.

Equipment shall operate reliably in the environmental conditions in which it is used.

8 Acceptance and rejection of requests and items

8.1 Acceptance and rejection of the customer's request

The forensic service provider shall have a documented procedure to review and record the customer's request which should include:

- a) criteria for acceptance or rejection of the customer's request;

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b) criteria for acceptance or rejection of items.

The criteria for accepting or rejecting a submitted request or submitted items for analysis shall be made available to the customers.

The request shall be reviewed for clarity to ensure the items are relevant to the request.

The customer's request for service shall be recorded.

The forensic service provider shall consult with the customer when the written request for service is ambiguous.

In instances where the request is rejected, the reason for the rejection shall be made available to the customer and the records should be retained.

8.2 Recording of accepted items

The acceptance of items shall be recorded. The rejection of items should be recorded. The reason for rejection of items should be recorded and be available to the customer.

The forensic service provider shall record any conditions regarding the packaging or integrity of items that can negatively affect the suitability for analysis. Where there is doubt about suitability of an item or when the item does not conform to the description provided, the forensic service provider should consult with the customer to clarify the issues and record the outcome.

Once accepted, if not already appropriately labelled and/or packaged, items shall be labelled, packaged and stored in accordance with ISO 21043-2. Where the integrity of an item has been compromised prior to analysis, measures shall be taken to preserve the item from further degradation, contamination, or alteration. The condition of the item and the measures taken shall be recorded.

9 Analytical strategy

9.1 General requirements

The analytical strategy shall take into consideration:

- a) prioritization of the items;
- b) task-relevant information;
- c) the nature, quality and relevance of the items;
- d) the capabilities and limitations of the available methods; and
- e) the appropriateness and benefits of the additional testing of items. In order to allow re-analysis when necessary, the analytical strategy should give consideration to retaining a portion of the item.

If the item will be fully consumed during analysis the customer should be notified prior to analysis. If the risk of fully consuming the item is always present for a type of analysis, and the customer has been informed about this risk, the request may be considered as implicit approval.

The forensic service provider shall have a documented procedure to address the common sequences of analysis.

9.2 Assessment prior to analysis

The forensic service provider shall assess whether or not analysis of the items would address the customer's request in the context of the information/case circumstances provided. A description of the items shall be recorded prior to their analysis.

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If the items are not sufficient to address the customer's request, the forensic service provider should consult with the customer regarding the limitations of the analysis given the items received. The forensic service provider should record the outcome of the consultation.

When analyses will be performed despite item suitability being questioned, this should be recorded.

9.3 Selection of methods

The selection of analytical methods shall be based on their suitability for the intended use and should consider:

- a) the purpose of the analysis (see [section 10](#));
- b) the nature of the item to be analyzed;
- c) the quantity of the item available for analysis;
- d) the method(s) available;
- e) the non-destructive or destructive nature of the method(s);
- f) the availability of suitable equipment and operators;
- g) the sensitivity and specificity;
- h) the accuracy required;
- i) the measurement uncertainty.

Where the quantity of an item is limited or the analysis is likely to consume or alter the item, the sequence of analysis shall be planned and implemented to ensure that non-destructive methods are given priority and conducted before any destructive methods are attempted.

Any deviation from a validated or verified analysis method shall be technically justified and recorded. The expected impact of the deviation on the validity of the method shall be recorded.

9.4 Item selection, sampling and preparation

The forensic service provider shall have documented procedures for item selection, sampling and preparation for each type of item that is routinely analyzed.

NOTE 1 Sampling of items is carried out for either or both of the following reasons:

- a) to obtain material in a form suitable for analysis;
- b) to obtain a representative sample of the item received.

Sampling methods shall take into account how representativeness is obtained when only part of an item is analyzed. When homogenization is necessary for the representativeness of the sample, but is not possible, several samples of the item should be analyzed separately.

NOTE 2 Recommendation for drug sampling strategies can be found in reference UN *Guidelines on Representative Drug Sampling*, United Nations Office on Drugs and Crime, ENFSI Drugs Working Group, 2011^[3].

10 Purpose of the analysis

10.1 Classification/Identification

The forensic service provider shall have a documented method for the comparison of items to relevant reference samples, collections and databases.

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Analysis for classification shall be based on the specificity of any features used and on the suitability of the database or the taxonomic key used.

Annex [B.1](#) includes examples of classification.

Methods that are used to identify classes of compounds and that have a high rate of false positive observations as a consequence of low specificity, shall be treated as presumptive only.

Analysis for identification shall use at least one method of high selectivity when available.

In the identification process, positive observations from methods of low specificity shall be confirmed using one or more methods of higher specificity, if methods of higher specificity are available.

Criteria required for identification shall be established for methods used.

NOTE Identification is a special case of classification.

Annex [B.2](#) includes examples of identification.

10.2 Quantification

Quantification methods shall have acceptance criteria that include, at minimum:

- a) control sample requirements;
- b) requirements for acceptable calibration.

The forensic service provider shall have a documented procedure to address when replicate quantitative testing is needed. When replicate testing is performed, at least two samples from the original item shall be analyzed, and the variation between the calculated quantities shall be within defined limits.

For any analytical method applied to determine the concentration or purity of a compound(s) of interest, the limits and the working range of the calibration curve shall be defined. Defined acceptance criteria shall require the quantitative value to be within the working range of the calibration curve and shall prescribe actions to take when a quantitative value is outside of this range.

If multiple observations from controls are determined to fall outside of the acceptance criteria, the method shall be checked, and the inconsistencies recorded and the record retained.

Annex [B.3](#) includes examples of quantification.

10.3 Question of source

Where analysis for question of source involves the comparison of an item from an unknown source with an item from a known source, the item from the known source shall be suitable for the required comparison, and shall be selected to be as representative as possible of the known source.

Annex [B.4](#) includes examples of questions of source.

10.4 Reconstruction

Analyses for reconstruction are carried out to address whether an observed effect could have been the consequence of a particular activity. Analyses for reconstruction should be based on empirical data and may involve experimental simulations under controlled conditions. If relevant to the case circumstances, such analyses shall consider whether alternative experimental conditions produce the same observed effect.

Annex [B.5](#) includes examples of reconstruction.

11 Reliability of observations

The forensic service provider shall have a documented procedure that includes checking of:

- a) analytical observations;
- b) data transfers;
- c) comparisons;
- d) measurements;
- e) manual calculations.

The procedure shall specify at what stage in analysis checks occur, who conducts checks, and what methods require checks to be performed by a second examiner competent in the same area of expertise. The forensic service provider shall have a documented procedure to address disputes arising from the check of analysis. A record of checks shall be retained.

Annex A **(informative)**

The forensic process

The figure below is a general scheme intended only to illustrate how the parts of the ISO 21043 series are related.

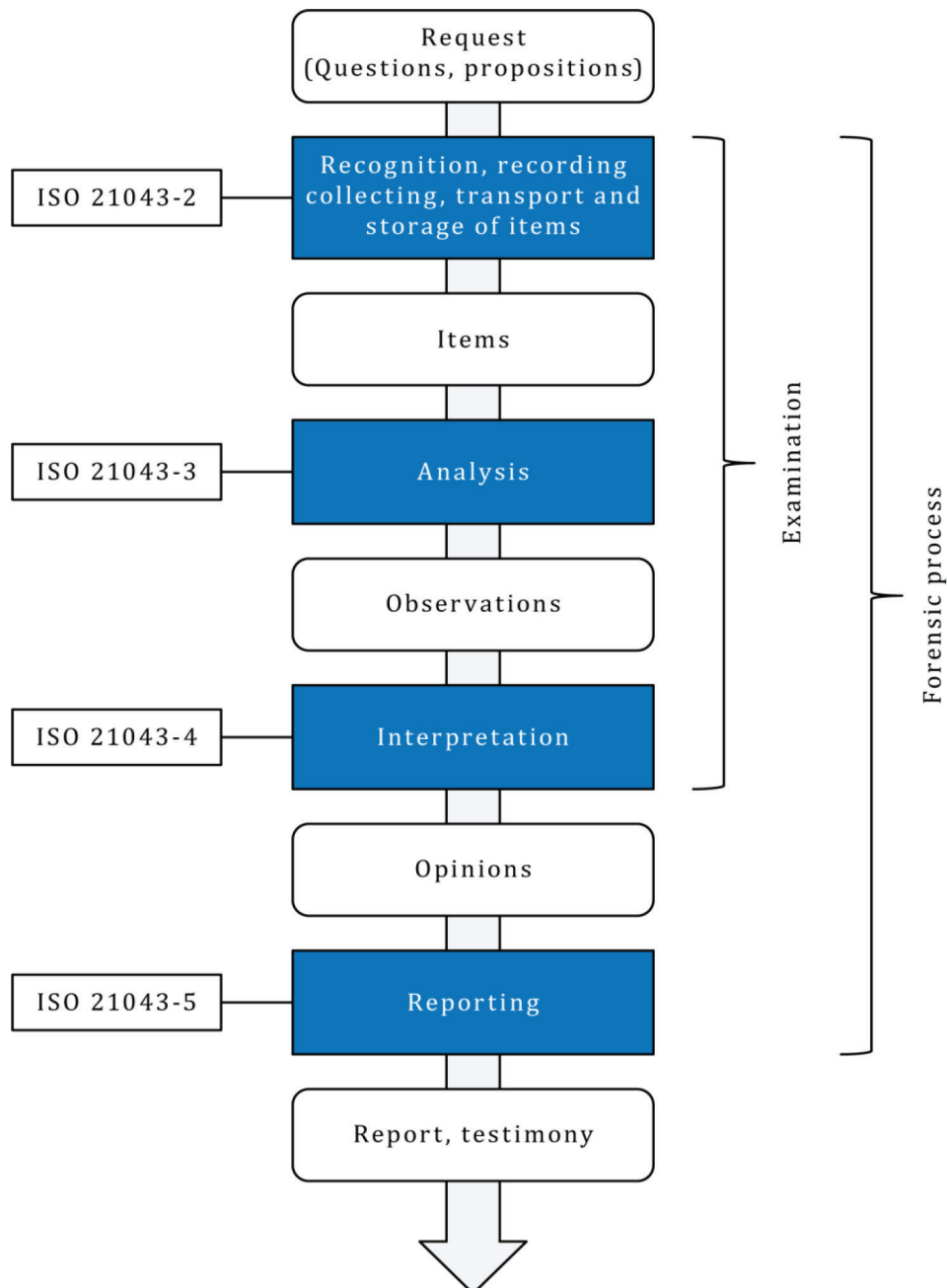


Figure A.1 — Forensic Process

The blue boxes indicate processes and are linked to the various parts of the standard (indicated with labels on the left side). The white boxes with rounded corners can be the input or output of a process.

The forensic process can be very diverse, and the arrow only illustrates the general flow. In a particular forensic process some actions might be skipped or repeated.

Annex B **(informative)**

Examples

B.1 Examples of classification

- EXAMPLE 1 Determining the manufacturer of a motor vehicle based on the comparison of a paint item from the crime scene with a motor vehicle paint database.
- EXAMPLE 2 Determining the manufacturer of a cartridge case, based on a comparison with a physical ammunition collection or a searchable database of cartridges from known manufacturers.
- EXAMPLE 3 Comparing a footwear mark (unknown) to a piece of footwear (known), to infer potential manufacturer.
- EXAMPLE 4 Analysing the properties of an impression and classify them as a finger, palm, or shoe impression.
- EXAMPLE 5 Comparing the properties of a bloodstain pattern of unknown type with the properties of bloodstain pattern of known type to determine the type of pattern.
- EXAMPLE 6 Comparing an unknown piece of malware found on a computer with a collection of pieces of malware representing known classes of malware.

B.2 Examples of identification

- EXAMPLE 1 Identifying human blood on an item through serological testing.
- EXAMPLE 2 In drug analysis, comparing the spectrum of an unknown item to that in a known drug reference database to identify the type of drug.
- EXAMPLE 3 Comparison between the RNA of an unknown type of cell present on an item with the RNAs from known types of cells.
- EXAMPLE 4 Identifying cannabis based on microscopic morphological properties.

B.3 Examples of quantification

- EXAMPLE 1 Determining the concentration of ethanol (alcohol) in blood.
- EXAMPLE 2 Measuring the quantity or the purity of cocaine in a white powder mixture.
- EXAMPLE 3 Determining the weight measurement of a suspected controlled substance.
- EXAMPLE 4 In digital casework, determining the number of illegal images in the total number of downloaded images.

B.4 Examples of questions of source

- EXAMPLE 1 Comparing a fingerprint found on an item (unknown source) to a reference fingerprint of a known person (known source).
- EXAMPLE 2 Comparing microscopic and chemical characteristics of carpet fibers recovered from the clothing of a dumped body (unknown source) to the interior carpet fiber in a suspect vehicle (known source).
- EXAMPLE 3 Comparing a footwear mark (unknown source) to a specific shoe (known source).
- EXAMPLE 4 Comparing the DNA found on an item (unknown source) with the DNA reference sample from a specific individual (known source).

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EXAMPLE 5 Comparing the handwriting on a document (unknown source) with reference samples of handwriting from a specific individual (known source).

EXAMPLE 6 Comparing the noise pattern from a digital picture (unknown source) with the noise pattern from digital pictures originating from the smartphone of a specific individual (known source).

B.5 Examples of reconstruction

EXAMPLE 1 Examining bloodstain patterns to determine possible activities that led to them.

EXAMPLE 2 Measuring features (e.g., position of cartridge cases, trajectories and impact locations of bullets) at a crime scene, to address the chain of events and activities leading to these observations.

EXAMPLE 3 Determining the presence of specific features in a file system, file sharing applications, and in digital images to address the chain of events leading to the presence of illegal images on a computer.

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DRAFT International Standard

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Forensic Sciences — Part 4: Interpretation

*Science criminalistique —
Partie 4: Interprétation*

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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

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For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT), see the following URL: www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 272, *Forensic Sciences*.

The following is a list of all parts in the ISO 21043 series:

ISO 21043-1, *Forensic Sciences – Part 1: Terms and definitions*

ISO 21043-2, *Forensic Sciences – Part 2: Recognition, recording, collection, transport and storage of items*

ISO 21043-3, *Forensic Sciences – Part 3: Analysis*

ISO 21043-4, *Forensic Sciences – Part 4: Interpretation*

ISO 21043-5, *Forensic Sciences – Part 5: Reporting*

Introduction

This document is part of a series of five that includes the different components of the forensic process from scene to courtroom. This document establishes requirements designed to safeguard the process for the interpretation of observations including the use of statistical models and/or human judgement to address alternative propositions based on the questions asked by the customer.

In this document, the following verbal forms are used:

- “shall” indicates a requirement;
- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or a capability.

Further details can be found in the ISO/IEC Directives, Part 2.

[Annexes A, B, C](#) and [D](#) are for information only.

The notes given provide clarification of the text, examples and guidance. They do not contain requirements.

Forensic Sciences —

Part 4: Interpretation

1 Scope

This document specifies requirements and provides recommendations for the interpretation of observations in order to reach opinions that answer questions that are relevant for decision making in investigations or legal proceedings. This document states requirements that are applicable to all forensic disciplines. This document states requirements that apply when the opinion is based directly on human judgment and when the opinion is based on interpretation through a statistical model. This document is applicable to interpretation that occurs at a scene, within a forensic facility or within a judicial setting.

Interpretation is not necessary, and the requirements of this standard do not apply, if the observations resulting from the analysis directly answer the relevant question.

EXAMPLE In analytical chemistry, substances are often identified or classified. Provided that the applied analytical methods are not limited in selectivity or sensitivity for the given question, the observations can lead to a direct statement of the name of the substance (identification) or a type of material (classification). This is not considered interpretation for the purposes of this standard.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 21043-1, *Forensic Sciences - Terms and definitions* apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>

4 General requirements

Forensic service providers shall have documented methods for interpretation.

The interpretation methods shall be suitable for the intended use and should be supported by relevant validation studies.

Any deviation from a documented interpretation method shall be technically justified and recorded.

Interpretation shall:

- a) be supported by task-relevant data, observations, and information that are within the examiner's area of expertise;
- b) be relevant to the question put forward by the customer;

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- c) take into account all available task-relevant information;
- d) be designed to reduce the risk of cognitive bias, including designed to attempt to limit the influence of information that is not task-relevant.

Examiners shall not interpret observations where that would take them outside their area of expertise.

The examiner shall record any observations, data, information, and interpretation methods used in forming an opinion.

Known limitations and sources of error in the analysis and interpretation methods used shall be recorded and taken into account in the interpretation.

5 Questions and propositions

5.1 Questions

5.1.1 General

The examiner should not address questions concerning the intent of an accused person.

The examiner shall not address questions defined in terms of guilt or innocence.

5.1.2 Types of questions

The types of questions asked in forensic science commonly include:

- a) what is the item? (classification/identification);
- b) how much is there of an item/how many are there of an item? (quantification);
- c) who or what is the source of an item/do two or more items have the same source? (question of source);
- d) what happened, where did it happen, when did it happen, who did what? (reconstruction).

5.1.3 Classification/Identification

The classes used for classification shall be clearly defined.

If classes are defined directly by observations that do not require interpretation, the requirements of this document do not apply to that classification.

Identification of an item shall be strictly separated from individualization, which refers to question of source (see [Section 5.1.5](#)).

5.1.4 Quantification

If quantitative data do not require interpretation, the requirements in this document do not apply.

Interpretation of quantitative data shall take into account measurement uncertainty.

If a sampling method is used, the interpretation shall consider the representativeness of the sample and how the sampling method impacts the opinion given.

NOTE Depending on the situation, quantification can require interpretation. For example, interpretation would be required when estimating the total number of items in a large number of containers when individual counting is not practicable. Conversely, if all the items were individually counted, then interpretation would not be required.

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5.1.5 Question of source

When considering the source of an item, the question is either:

- a) whether the questioned-source item is from the same source as the known-source item(s) or from a different source, or
- b) whether multiple items come from the same source or from different sources.

NOTE 1 The source of the item does not imply the activity by which the item was transferred.

The interpretation of a comparison of items shall take into account the rarity (with respect to the relevant population) of the observed features of the items or the observed comparison scores.

It shall also take into account the following sources of variation with respect to observations made on questioned and known-source items:

- c) within-source variability: the variability in properties of different items from the same source;
- d) between-source variability: the variability in properties of items from different sources;
- e) measurement variability: the variability in repeated observations of the same item.

NOTE 2 It is possible for items of different sources to be similar.

5.1.6 Reconstruction

5.1.6.1 Activity

If addressing questions about activity in evaluative interpretation, for each proposition considered, the examiner shall, when appropriate, take into account the probability, if that proposition were true, that:

- a) the item(s) would be transferred from its alleged point of origin to its current location (transfer probability);
- b) the item(s) would persist in its current location given the time elapsed since it was allegedly transferred and relevant events that have occurred since that time (persistence probability);
- c) the item(s) would be recovered from its current location (recovery probability);
- d) the item(s) would be present at its current location for reasons unrelated to the alleged activity (background probability).

5.1.6.2 Event reconstruction

If conducting event reconstruction, the examiner shall consider one or more alternative propositions or alternative explanations for the observations, addressing issues such as:

- a) What activities and events occurred?
- b) Where did they occur?
- c) When did they occur?
- d) In what order did they occur (timeline)?
- e) Who carried out the activities?

5.2 Addressing the question with either investigative or evaluative interpretation

5.2.1 General

Examiners may address questions using either investigative or evaluative interpretation.

Investigative interpretation can generate likely explanations of observations, can test these explanations with new observations, and can eliminate or re-rank the explanations.

Investigative interpretation can also lead to an estimation that explains the observations.

Evaluative interpretation should be used if a relevant alternative proposition can be formulated.

EXAMPLE 1 If the question were what speed a car was moving, the examiner would use investigative interpretation to give an estimate of the speed. If the question were whether the speed was above the speed limit or not, the examiner would use evaluative interpretation.

EXAMPLE 2 If the question were about the time since death, the examiner would use investigative interpretation to give an estimate of the time since death. If the issue were whether or not the victim died within the timeframe for which the accused has an alibi, the examiner would use evaluative interpretation.

5.2.2 Investigative interpretation

In investigative interpretation, the examiner will not have propositions at the outset.

The examiner may generate possible explanations for the observations. Those explanations may be turned into propositions for a later evaluative interpretation.

The examiner may also provide estimations which explain the observations. Estimation of a value does not involve the use of propositions.

5.2.3 Evaluative interpretation

When propositions are used to address case-relevant questions, they shall be specified before the analysis is conducted. Routinely used propositions may be specified in a relevant procedure. If propositions are changed after the analysis has been conducted, such changes shall be justified and recorded.

The examiner shall consider at least two competing propositions which shall be:

- a) relevant to answering the question posed in the case;
- b) mutually exclusive;
- c) explicit and specific;
- d) considered during analysis, interpretation, and reporting.

The propositions should be provided by the customer, agreed on by the customer, or follow directly from their request.

6 Types of opinions

6.1 General

Various types of opinions can be appropriate, depending on the questions asked, the purpose of the opinion, and the information available.

The types of opinions addressed in the next subsections are:

- a) relative probabilities of the observations given the propositions (likelihood ratios);
- b) categorical opinions on the propositions;

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- c) generated explanations for the observations;
- d) estimations.

NOTE 1 Probabilities can be assigned quantitatively using statistical models, or quantitatively or qualitatively based on professional judgment. The quantitative and qualitative assignment of probabilities is addressed in [Section 7](#).

NOTE 2 The quantitative and qualitative expression of opinions is addressed in [Sections 8.2](#) and [8.3](#), respectively.

6.2 Likelihood ratios

When assigning a likelihood ratio, the examiner shall assess the ratio of the probabilities of the observations if the first proposition were true versus if the second proposition were true. If more propositions are considered, the examiner shall assess the probability of the observations under each pair of propositions and assign their ratios.

The value of a likelihood ratio may be assigned and expressed quantitatively or qualitatively.

The examiner shall not confuse the likelihood ratio with the ratio of the probabilities of the propositions.

NOTE 1 The ratio of the probabilities of the propositions, given the observations, is known as the “posterior odds”. The likelihood ratio is not the same as the posterior odds. Confusing a likelihood ratio with the posterior odds is known as the fallacy of the transposed conditional, also referred to as the “prosecutor’s fallacy”.

See [Annex B](#) for further information about likelihood ratios.

NOTE 2 A likelihood ratio can be assigned quantitatively using statistical models, or quantitatively or qualitatively based on professional judgment. The quantitative and qualitative assignment of probabilities is addressed in [Section 7](#).

6.3 Categorical opinions

Categorical opinions shall only be formed when, in the examiner’s judgement, it would be impossible to obtain the observations if any alternative proposition were true.

6.4 Explanations

When conducting an investigative interpretation, an examiner may generate explanations.

The examiner may rank the explanations by their probabilities.

When assigning probabilities of explanations, the examiner shall take into account not only the current observations but all available relevant observations and information in the case. When evaluating the probabilities of the explanations, the examiner shall take into account all explanations considered, all observations obtained from all relevant analyses, and any relevant case information.

6.5 Estimations

In investigative interpretation, an examiner may estimate a value. The examiner should assess the uncertainty of the estimation, e.g., in the form of a distribution, or a range of values.

7 Assigning probabilities

7.1 General

Probabilities are assigned using observations, existing data, and/or statistical models. Subjective assignments of probabilities may be used in the absence of this empirical information. Subjective probabilities should be clearly expressed.

The examiner shall document how probabilities are assigned, including the basis for such assignments. The examiner shall be transparent about the sources of information used to assign probabilities.

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Any assumptions used to assign probabilities should be clearly recorded.

Data used for assigning probabilities shall be as representative as possible of the relevant population of items and as representative as possible of the conditions of the case.

7.2 Statistical-model-based assignment of probabilities

Statistical models and data used to assign probabilities shall be suitable for their intended use.

Data shall only be used for assigning probabilities if, in the examiner's judgement, they are sufficiently representative of the relevant population and the conditions of the case.

The data used to assign probabilities should be from published sources or from unpublished sources that can be made available. If the data used to assign probabilities are from unpublished sources, they should be made available upon request.

7.3 Human-based assignment of probabilities

When assigning quantitative or qualitative probabilities, examiners shall use professional judgment in conjunction with relevant observations, relevant information, and relevant data.

8 Expression of the opinion

8.1 General

The opinion and the basis for the opinion shall be recorded.

The opinion may be expressed as:

- a) a quantitative or qualitative likelihood ratio;
- b) a quantitative or qualitative probability of the observations under each proposition (when more than two propositions are considered);
- c) a categorical opinion on the propositions;
- d) generated explanations of the observations;
- e) an estimation.

The opinion expressed shall explicitly refer to any propositions considered.

If no opinion can be expressed, the reasons shall be recorded.

All opinions involve professional judgement and are not statements of fact.

8.2 Quantitative expression of opinion

If a numerical value (e.g., a probability or a likelihood ratio) is calculated using quantitative observations, quantitative data, and a statistical model, the opinion should be expressed using that numerical value.

Estimations shall be expressed numerically and should be accompanied by an expression of the uncertainty of the estimate.

8.3 Qualitative expression of opinion

Qualitative opinions shall be drawn from a predefined opinion scale on which each level has a verbal expression. Examples of opinion scales are given in [Annex C](#).

The verbal expressions on the opinion scale should be defined using ranges of numerical values.

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If the interpretation method does not produce a numerical value, a verbal expression shall be recorded. The opinion scale should not be used to convert a verbal expression to a numerical value.

8.4 Categorical opinions

Categorical opinions shall not be expressed as if they were statements of fact.

Categorical opinions should be checked. This check should include examination of the items involved.

Forensic service providers shall determine when to perform a check of categorical opinions based on risk assessment. If input is received from customers, it shall be taken into account when considering risk. Records of the risk assessments shall be retained.

NOTE The risk-based decision to determine the proportion of cases requiring a check of categorical opinions based on human judgment can include factors such as the potential for an error, the impact of the types of reporting errors and experience of the examiner. The proportion of cases that require a check of categorical opinions based on human judgment can be different within and between disciplines.

The checks shall be performed by a second competent examiner in the same area of expertise and shall be recorded.

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Annex A (informative)

The forensic process

In [Figure A.1](#) below, is a general scheme intended only to illustrate how the parts of the ISO 21043 series are related.

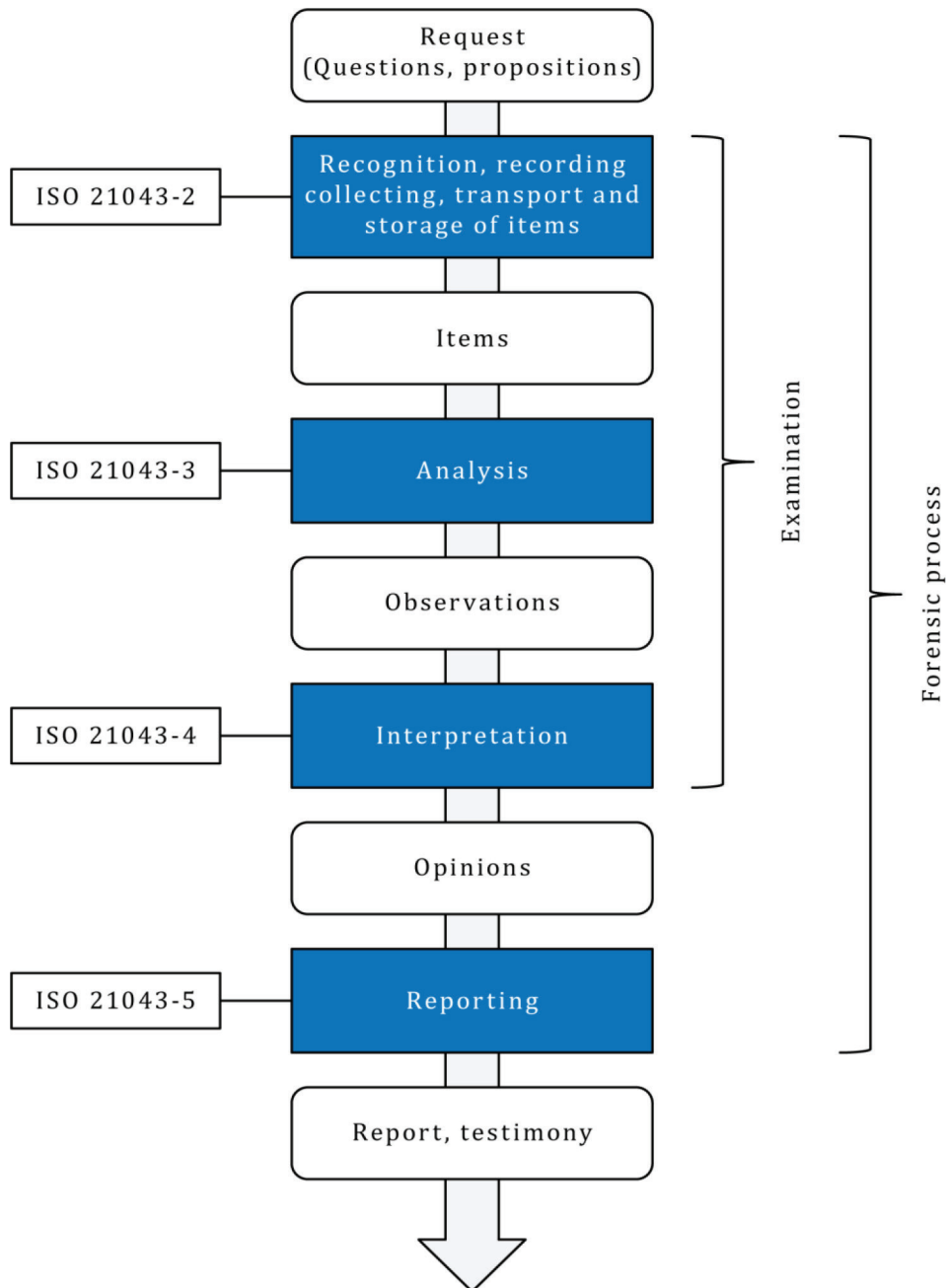


Figure A.1 — Forensic Process

The blue boxes indicate processes and are linked to the various parts of the standard (indicated with labels on the left side). The white boxes with rounded corners can be the input or output of a process.

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The forensic process can be very diverse, and the arrow only illustrates the general flow. In a particular forensic process some actions might be skipped or repeated.

Annex B **(informative)**

The meaning of a likelihood ratio

This annex is provided in order to explain the meaning of a likelihood ratio. It is not intended to instruct a court of law how to reason. In the text below, a “decision maker” is anyone who wants to make use of a likelihood ratio presented by an examiner. For the purpose of explaining the meaning of a likelihood ratio, we assume that the decision maker accepts the likelihood ratio presented by the examiner and makes use of that likelihood ratio.

An examiner uses a likelihood ratio as an expression of their assessment of the relative probabilities of the observations if one of two competing propositions were true versus if the other proposition were true. Hereinafter these will be referred to as proposition *A* and proposition *B*.

A likelihood ratio equals the probability of the observations if *A* were true divided by the probability of the observations if *B* were true. The purpose of a likelihood ratio is to express the probative value of the observations supporting proposition *A* or *B*. If a decision maker accepts the likelihood ratio presented by the examiner, that likelihood ratio expresses the amount by which the decision maker would, in light of the observations, update their belief as to the relative probability that *A* is true versus that *B* is true. This is formally expressed by Bayes’ Theorem as:

The prior odds multiplied by the likelihood ratio equals the posterior odds.

The prior odds quantify the decision maker’s belief that *A* is true divided by their belief that *B* is true *before* the examiner presents their opinion. The likelihood ratio quantifies the impact of the observations on the odds of propositions *A* and *B*. The posterior odds quantify the decision maker’s belief that *A* is true divided by their belief that *B* is true *after* taking into account the examiner’s opinion.

NOTE Bayes’ Theorem is a statement of logic. It is not a theory. It can be proven from the basic laws of probability.

The decision maker considers all the evidence presented in the case. In contrast, the likelihood ratio presented by the examiner is based only on the observations within the examiner’s area of expertise and on task-relevant information, it does not depend on other evidence.

For more comprehensive introductions to the likelihood-ratio framework in general, see [1-4]. For introductions contextualized in different branches of forensic science, see, for example: for DNA[5, 6], for voice recordings[7], and for physicochemical data[8].

Annex C (informative)

Examples of opinion scales

Introduction

The following are informative examples only. Their provision here does not imply that use of any of these particular scales is either required or recommended.

Example 1

The following is an example of an opinion scale, i.e., ordinally ranked verbal expressions for describing different degrees of strength of opinion. This example is based on the ‘ENFSI guideline for evaluative reporting in forensic science’^[9]. It provides verbal expressions that describe likelihood ratios, and the meaning of each level in the scale is defined using a range of numeric likelihood ratio values.

A likelihood ratio expresses an examiner’s assessment of the relative probabilities of the observations if one of two competing propositions were true versus if the other proposition were true. In the [Table C.1](#) below, these propositions are referred to as proposition *A* and proposition *B*.

Table C.1 — Opinion scale example 1

Range of numeric likelihood ratio values	Verbal expression of strength of opinion
1 – 2	The observations are <i>approximately equally probable</i> irrespective of whether proposition <i>A</i> were true or proposition <i>B</i> were true. The observations <i>do not support</i> one proposition over the other.
2 – 10	The observations are <i>slightly more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>weak support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
10 – 100	The observations are <i>more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>moderate support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
100 – 1,000	The observations are <i>appreciably more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>moderately strong support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
1,000 – 10,000	The observations are <i>much more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>strong support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
10,000 – 1,000,000	The observations are <i>far more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>very strong support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
1,000,000 or more	The observations are <i>exceedingly more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>extremely strong support</i> for proposition <i>A</i> relative to proposition <i>B</i> .

If the observations were more probable if proposition *B* were true than if proposition *A* were true, then in the example above, the order of proposition *A* and proposition *B* in the verbal expressions would be reversed.

Example 2

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[Table C.2](#) gives another example of a scale with verbal expressions for strength of opinion, applied to a question of source.

Table C.2 — Opinion scale example 2

Verbal expression	Description
Strongest support for same source	A lot of similarity and no unexplainable difference could be observed in the comparison. The examiner would not expect to make the observations in an item from a different source.
Strong support for same source	Similarity and no unexplainable difference could be observed in the comparison. The examiner has a much higher expectation of making the observations in an item from the same source than in an item from a different source.
Limited support for same source	A little similarity and no unexplainable difference could be observed in the comparison. The examiner has a somewhat higher expectation of making the observations in an item from the same source than in an item from a different source.
No support	The observations do not provide support for one proposition over the other. The examiner's expectancy for the observations of the comparison is similar for an item from the same or from a different source. This expression is not meant to be used when no comparison could be carried out.
Limited support for different source	A little difference and no specific similarity could be observed in the comparison. The examiner has a somewhat higher expectation of making the observations in an item from a different source than in an item from the same source.
Strong support for different source	Difference and no specific similarity could be observed in the comparison. The examiner has a much higher expectation of making the observations in an item from a different source than in an item from the same source.
Strongest support for different source	A lot of difference (or any unexplainable difference) and no specific similarity could be observed in the comparison. The examiner would not expect to make the observations in an item from the same source.

Further reading

For additional examples of opinion scales, see [\[9-12\]](#).

Annex D **(informative)**

Examples of the application of the Interpretation standard

This annex aims to illustrate application of the Interpretation standard in casework.

The following are informative examples only. They do not imply that the use of a particular wording or scale is required or recommended, and other wording that adheres to the requirements can be used. Additionally, the examples do not include direction on how the interpretation should be reported.

For brevity, some underlying assumptions and the full scale from which the verbal expressions were chosen were left out.

Handwriting example (question of source, evaluative interpretation, qualitative likelihood ratio)

In a fraud case, the question was whether a questioned signature was authentic. Reference signatures were obtained for comparison by a handwriting examiner. The propositions are: (Proposition *A*) the questioned signature was written by the same person as the reference signatures, and (Proposition *B*) the questioned signature was written by some other person from the relevant population. The comparison of the signatures resulted in a number of observations, among which a few similarities, and differences which were within the natural variation of signatures. The examiner assessed the probability of the observations for both propositions, and expressed their opinion as: “my observations provide limited support for Proposition *A* relative to Proposition *B*”.

Forensic medicine example (reconstruction, evaluative interpretation, qualitative likelihood ratio)

In a possible child abuse case, Person X is accused of having caused head trauma to a child while the child was in Person X's care. Magnetic-resonance imaging (MRI) scans were taken of the child's head the day after the alleged abuse. Person X denies that the head trauma was caused by that person, and proposes that the head trauma was caused 2 weeks earlier when the child fell from their bicycle. The fact that the child fell from the bicycle two weeks earlier is not disputed. The propositions are: (Proposition *A*) The head trauma was caused the day before the MRI scans were taken, and (Proposition *B*) the head trauma was caused 2 weeks ago. A radiologist examined the MRI scans. The radiologist observed a mixed intensity lucent area in the left frontal area on the MRI scans. The radiologist reported that “observations are much more probable if the child had suffered bleeding due to the head trauma 2 weeks before the MRI scans were taken than if the child had suffered bleeding due to head trauma one day before the MRI scans were taken.”

Fingerprinting example (question of source, evaluative interpretation, categorical opinion)

In a burglary case, a fingerprint was found at the crime scene, and the question was whether it originated from Person X. The fingerprints of Person X were taken for comparison with the fingerprint. The propositions are: (Proposition *A*) the fingerprint came from Person X, and (Proposition *B*) the fingerprint came from some other person in the relevant population. The fingerprint examiner found many differences between the fingerprint and the friction ridge detail in Person X's fingerprints. In the examiner's judgement these observations would be impossible if Person X was the source of the fingerprint found at the crime scene. The examiner phrased their opinion as: “In my opinion, Person X was not the source of the fingerprint. I found many differences between the friction ridge detail in the fingerprint and the friction ridge detail in Person X's fingerprints. In my opinion, it would be impossible to observe these differences if the fingerprint originated from Person X.”

Voice comparison example (question of source, evaluative interpretation, quantitative likelihood ratio)

In a fraud case, the question was whether Person X was the person speaking on a telephone-call recording. Recordings of telephone calls in which the speaker was known to be Person X were also available. The speakers of interest in the questioned-speaker and known-speaker recordings were all assumed to be adult females speaking the same language with the same accent. The propositions were: (Proposition *A*)

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The questioned-speaker recording and the known-speaker recordings are recordings of the same speaker, and (Proposition *B*) the questioned-speaker recording and the known-speaker recordings are recordings of two different speakers from the relevant population. The relevant population was defined as adult female speakers of the same language and accent as spoken on the recordings.

The examiner used a forensic-voice-comparison system based on automatic-speaker-recognition technology. The system analyzed acoustic properties of the voices on the recording, and entered the resulting observations into statistical models that calculated likelihood ratios. The examiner calibrated and validated the system using recordings of speakers sampled from the relevant population and recorded under conditions that reflected those of the questioned-speaker and known-speaker recordings.

The examiner reported that “the probability of obtaining the acoustic properties of the questioned-speaker and known-speaker recordings is 100 times greater if they were recordings of the same speaker than if they were recordings of two different speakers from the relevant population.”

Fiber examination example (reconstruction, evaluative interpretation, qualitative likelihood ratio)

In an assault case against Person X, the prosecution alleges that: (Proposition *A*) Person X hit the alleged victim, while the defense claims that (Proposition *B*) Person X did not hit Person Y. A fiber examiner is asked to examine the clothing of Person X and the clothing of Person Y for fiber evidence. It is not disputed that, before the alleged assault, Person X and Person Y greeted and hugged each other.

In view of these agreed case circumstances and the observed absence of fibers due to the low fiber retention of the jackets that Person X and Person Y were wearing, the examiner concludes that: “The observations are approximately equally probable, irrespective of whether Proposition *A* were true or Proposition *B* were true. The observations do not support one proposition over the other.”

DNA analysis example (question of source, evaluative interpretation, quantitative likelihood ratio)

A bloodstain was found at a crime scene, and the question was if Person X was the source of the DNA in the item. The propositions are: (Proposition *A*) The DNA came from Person X, and (Proposition *B*) The DNA came from some other, unrelated person. The examiner expressed their opinion as “the observed DNA profile is at least one billion times more probable if the DNA came from Person X than if the DNA came from some other, unrelated person.”

Fire investigation example (classification, investigative interpretation)

After a building burned down, the police consider whether a criminal investigation should be started, and ask an examiner about the cause of the fire. After examining the scene of the fire, the examiner sees three possible explanations for the fire: (Explanation *A*) an electrical malfunction, (Explanation *B*) an overheated stove while cooking, and (Explanation *C*) adding a source of ignition by a person. For the purpose of helping the police decide how to proceed, the examiner ranked the explanations for the fire by their probability. The examiner expressed the investigative opinion by stating that “based on my observations and not knowing anything else about the case, in my opinion Explanation *A* is the most probable, Explanation *C* is much less probable, and the probability of Explanation *B* is very low.”

Image analysis example (estimation, investigative interpretation)

The police want to know the speed of a car that is visible in a security camera's footage. The examiner has analyzed the frame rate of the camera that was involved, and estimated the positions of the car in the frames in which it is visible. Based on the timing and the estimated distance travelled between the frames, the examiner offered the following opinion on the speed: “based on the analyzed images and the known frame rate of the camera, my estimation of the speed of the car is 72 ± 5 km/h”.

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Forensic Sciences — Part 5: Reporting

*Science criminalistique —
Partie 5: Rapports*

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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see the following URL: www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 272, *Forensic Sciences*.

The following is a list of all parts in the ISO 21043 series:

ISO 21043-1, *Forensic Sciences – Part 1: Terms and definitions*

ISO 21043- 2, *Forensic Sciences – Part 2: Recognition, recording, collection, transport and storage of items*

ISO 21043- 3, *Forensic Sciences – Part 3: Analysis*

ISO 21043- 4, *Forensic Sciences – Part 4: Interpretation*

ISO 21043-5, *Forensic Sciences – Part 5: Reporting*

Introduction

This document is part of a series of five that includes the different components of the forensic process from scene to courtroom. This document establishes requirements to safeguard the process for reporting.

In this document, the following verbal forms are used:

- “shall” indicates a requirement;
- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or a capability.

Further details can be found in the ISO/IEC Directives, Part 2.

[Annexes A](#) and [B](#) are for information only.

The notes given provide clarification of the text, examples and guidance. They do not contain requirements.

Forensic Sciences —

Part 5: Reporting

1 Scope

This document specifies requirements and provides recommendations for reporting of forensic work. It is applicable to the reporting of forensic work performed at a scene and within a forensic facility. The requirements facilitate the preparation and issuing of reports that are accurate, clear, transparent, complete, unambiguous, impartial and suitable for their intended use. This document also includes requirements relating to the review of case records and reports, the issuance and control of reports, and requirements related to testimony.

NOTE Legal requirements take precedence over any requirements in this standard.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 21043-4, *Forensic Sciences Part 4: Interpretation*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 21043-1, *Forensic Sciences Part 1: Terms and definitions* apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>

4 General requirements

Reports shall be accurate, clear, transparent, complete, unambiguous, impartial, and suitable for their intended use.

The forensic service provider shall have a documented procedure for formally authorizing examiners to issue reports.

Examiners shall not report beyond their area of expertise.

Examiners shall not report beyond what can be based on the available information.

Known limitations of methods, procedures, observations, and opinions, should be stated clearly in the report or in an appendix attached to the report. If the limitations are not stated in the report or in an appendix to the report it shall be stated that the information on the limitations is available and shall be provided upon request.

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Descriptions of any deviation from documented methods or procedures shall be included in the report if the deviation could have a substantial impact on the observations or opinions. The reason for any such deviation shall be explained in the report or in an appendix attached to the report.

Any terms that are not in common usage, or for which the specialist usage differs from the common use, should be clearly defined and communicated to the customer (e.g., in a glossary appended to the report).

[Annex B](#) provides information on different types of reports.

5 Report contents

5.1 General

All reports shall include:

- a) a description of the items examined including their unique identifiers;
- b) a description of the examinations undertaken;
- c) the results of examinations, including observations (as relevant) and opinions reached;
- d) an identifier of the person(s) responsible for the content of the report.

All reports should also include or reference:

- e) what was requested;
- f) who made the request;
- g) when the request was made and/or received;
- h) date items were collected or received;
- i) identifying information for scenes examined, if applicable;
- j) method(s) used;
- k) additional information relied upon during the examination (e.g., photographs, other forensic reports, task-relevant circumstances of the case that were considered);
- l) a list of the items accepted for analysis but not analysed;
- m) details of the environmental conditions under which the examinations were conducted where such conditions lie outside the expected operating range and could have affected the results of the examination;
- n) if analysis has been performed despite the suitability of the item for analysis having been questioned by the forensic service provider;
- o) a signature or e-signature of the person(s) responsible for the content of the report.

5.2 Administrative requirements

All reports shall include:

- a) a title that clearly designates the document as a report (e.g., by including the word "report" in the title);
- b) a means of uniquely identifying the report;
- c) the date of issue;
- d) a unique case identifier (e.g., case number);

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- e) the name of the forensic service provider;
- f) pagination (e.g., page 1 of 10);
- g) a means of signifying the end of the report;
- h) reference to any appendices or attachments.

NOTE to a): Reports are sometimes referred to as statements, certificates, or expert reports.

5.3 Item integrity

When there is an indication that the integrity of an item has been compromised prior to analysis, the description of the item in the report shall include details of the noticed irregularities such as the packaging and condition of the item as received.

The report shall also describe any mitigating actions taken.

5.4 Significant figures and measurement uncertainty

When quantitative observations are reported, the number of significant figures used to report the final values shall be supported by the measurement uncertainty of the method used.

Measurement uncertainty shall be reported where relevant and applicable.

5.5 Reporting opinions

5.5.1 General

If opinions are reported, the opinions shall conform to ISO 21043-4.

The report shall differentiate opinions from observations.

Reports that contain opinions shall include:

- a) the questions to be answered;
- b) the set of propositions, where applicable;
- c) a description of assumptions that are relevant for forming the opinion;
- d) the basis of the opinion.

If reports contain opinions and an opinion scale is used, a reproduction of the entire scale shall be included or referenced in the report.

Reports that contain opinions should include or reference:

- e) a description of any data used;
- f) a description of any statistical models used.

Opinions shall not be presented in such a way as to overstate or understate their strength.

Where the strength of the opinion is reported as a likelihood ratio, a bound of the likelihood ratio closer to the neutral value of 1 may be reported, making clear that the likelihood ratio exceeds that bound. The forensic service provider shall have a documented rationale for choosing that bound.

EXAMPLE 1 In a DNA profile comparison, an opinion could state that “the likelihood ratio is greater than 1 billion”. This would not be understating the strength of the opinion.

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NOTE Lowering a likelihood ratio based on limitations of the quality of the item, measurement or data, is not understating the strength of the opinion, but is following the requirements to take limitations into account in the interpretation, refer to ISO 21043-4.

Opinions, including categorical opinions, shall not be presented as if they were statements of fact.

When reporting opinions based on questions of source, the examiner should explicitly state that only the source of the item was considered, and not the activity through which the item was transferred.

5.5.2 Reporting probabilities

If probabilities are assigned using data, the data used should be described and/or referenced in the report.

If a method used for assigning probabilities produces a numerical value, the opinion should be expressed using that numerical value.

If probabilities are assigned using the examiner's judgment, the basis for that judgment should be described in the report.

5.5.3 Reporting likelihood ratios

When reporting a likelihood ratio the examiner:

- a) shall express the likelihood ratio in a way that makes it clear that it is the ratio of the probabilities of the observations given the competing propositions; and
- b) shall not transpose the conditional.

NOTE The ratio of the probabilities of the propositions, given the observations, is known as the "posterior odds". The likelihood ratio is not the same as the posterior odds. Confusing a likelihood ratio with the posterior odds is known as the fallacy of the transposed conditional, also referred to as the "prosecutor's fallacy".

The likelihood ratio may be expressed quantitatively and/or qualitatively.

If the interpretation method calculated a quantitative likelihood-ratio value, that quantitative value should be reported.

5.5.4 Reporting estimations

Estimations shall be reported numerically and should be accompanied by an expression of the uncertainty of the estimate.

6 Case file review

6.1 General

When undertaken, case file review should be performed prior to the issue of the report to the customer.

Case file review consists of one or both of the following:

- a) technical review,
- b) administrative review.

The technical and administrative reviews may be performed separately or as one process.

When undertaken, case file review may be limited to the part of the case file related to the examiner and their area of expertise:

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Within that part of the case file, case file review shall include an assessment of records related to recognition, recording and collecting of items, analysis of items, interpretation of observations, and the reporting of observations and opinions made by the forensic service provider.

The type of review, parts of the case file reviewed, outcome of the review, date completed, and the identity of the reviewer shall be recorded, and the records shall be retained.

6.2 Technical review

Forensic service providers shall determine when to perform technical reviews based on risk assessment and shall have documented procedures for technical reviews including:

- a) the type and proportion of case files that require technical review;
- b) how technical reviews of case files are performed;
- c) the process to be followed when examiner and reviewer disagree regarding the analysis performed, any interpretation, or reported observations or opinions.

Reports that include opinions directly based on human judgement present a higher risk. This higher risk shall be considered in the risk assessment.

Records of the risk assessments shall be retained.

NOTE 1 The risk-based decision to determine the type and proportion of cases requiring a technical review can include factors such as the type of results being reported, the potential for an error, the impact of the types of reporting errors and experience of the examiner. The proportion of cases that require a technical review can be different within and between disciplines.

Technical review(s) shall be performed by an examiner who is competent to review in the relevant area of expertise, but shall not be performed by the examiner who carried out the examination.

When technical review is required by the procedure but is intentionally not performed (e.g., as a report is required urgently by the customer), the report shall contain a disclaimer stating that the review was not completed and why it was not completed. In these circumstances, the preliminary report should be followed by a final report that has been subject to a technical review, indicating any changes as an outcome of the review.

Disagreement on opinions between a reviewer and examiner that cannot be resolved shall be recorded and retained and should be included in the report if the report includes the disputed opinion.

NOTE 2 Information about the disagreement may be limited to the factual information that there is disagreement without further explanation.

6.3 Administrative review

Forensic service providers shall determine when to perform administrative reviews based on risk assessment and shall have documented procedures for administrative review.

Administrative review(s) shall be performed by a person who is familiar with the forensic service provider's operational requirements, and who has been deemed competent to carry out the review. The review shall not be performed by the examiner who carried out the examination.

7 Issue and control of reports

7.1 Issuing reports

Forensic service providers shall have documented procedures for the issuing of reports. The procedures shall address the following:

- a) the general circumstances under which reports are issued or not issued;

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- b) the types of reports that may be issued, including:
 - i) the circumstances under which each type is issued;
 - ii) what competence is required to issue reports of each type;
 - iii) the circumstances when a caveat is required (e.g., if a report is released prior to technical review);
 - iv) the required format and content;
- c) which reports are to be reviewed prior to issue.

If a request is accepted, a report should be issued.

7.2 Supplementing, amending and withdrawing reports

Forensic service providers shall have documented procedures for issuing supplementary or amended reports, and for withdrawal of reports. The procedures shall include the following requirements:

- a) customers shall be notified of a supplementary, amended or withdrawn report, and the reasons for it;
- b) supplementary reports shall reference the report that was supplemented,
- c) Amended reports shall clearly identify what has changed or has been added to the report;
- d) reports shall have a unique identifier that distinguishes between reports that are withdrawn and those that are replaced.

Customers shall be provided with a copy of any supplementary or amended report.

7.3 Controlling reports

Copies of all reports, including amended, supplementary and withdrawn, shall be retained as part of the case file.

The issued report shall be traceable so that any alterations can be identified by comparing to retained copies.

All issued reports shall undergo record control.

When reports are transmitted electronically, they shall be issued in a format so as to minimize the risk of unauthorized alteration.

7.4 Communications of observations and/or opinions other than by way of a report

In the course of an investigation, it can be necessary to provide observations and/or opinions to the customer in a timely manner. This information may be provided in a format agreed upon between the customer and the forensic service provider. When provided, the information shall meet the following requirements:

- a) opinions or observations that would require a review as described in [6.2](#) shall either:
 - i) be reviewed prior to release or
 - ii) accompanied by a caveat indicating that review has not been conducted;
- b) substantial limitations of the observations and/or opinions and their preliminary nature shall be stated;
- c) a record of the communication shall be retained and included in the case file and shall include:
 - i) a copy or summary of the information provided and a reference to the case,
 - ii) the date the information was provided,

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- iii) sufficient detail to determine how the information was provided, who provided the information and who the information was provided to.

In addition, observations should be checked before they are communicated.

These requirements apply irrespective of whether the information is communicated orally or in writing.

When the information is intended to be used for court, a report meeting all requirements of this standard should be issued to the customer.

8 Testimony

8.1 General

Testimony shall be accurate, clearly stated, responsive to the questions asked, impartial, and within the examiner's area of expertise.

Opinions shall not be presented in such a way as to overstate or understate their strength.

Opinions, including categorical opinions, shall not be presented as if they were statements of fact.

The expert witness should use terminology in common usage. If the usage differs from the common use, it should be clearly defined and communicated to the legal forum.

Opinions should be clearly distinguished from observations.

Observations and opinions should be supported by reports, case records, data, and/or documents.

8.2 Testimony training and evaluation

If personnel provide testimony, the forensic service provider shall have a documented procedure for testimony training, and shall have a documented procedure for assessment of competence of personnel to provide testimony.

If a testimony review is to be performed, the procedure should include:

- a) the frequency of review;
- b) who is permitted to perform the review;
- c) how the review is recorded, including identity of the reviewer and the date(s) performed;
- d) the process whereby feedback is given to the personnel providing testimony.

The assessment of competence to provide testimony should assess whether:

- e) the strength of the opinion was not overstated or understated;
- f) the basis for opinions was clearly described;
- g) clarifications and qualifications were provided (e.g., when affirmative or negative answers are requested, but the answer may be misleading);
- h) the testimony was within the area of expertise of the personnel providing testimony;
- i) the personnel providing testimony used terminology that was appropriate to the legal forum;
- j) the testimony was presented in an impartial manner, not misleading, and responsive to the questions asked.

Annex A (informative)

The forensic process

In [Figure A.1](#) below is a general scheme intended only to illustrate how the parts of the ISO 21043 series are related.

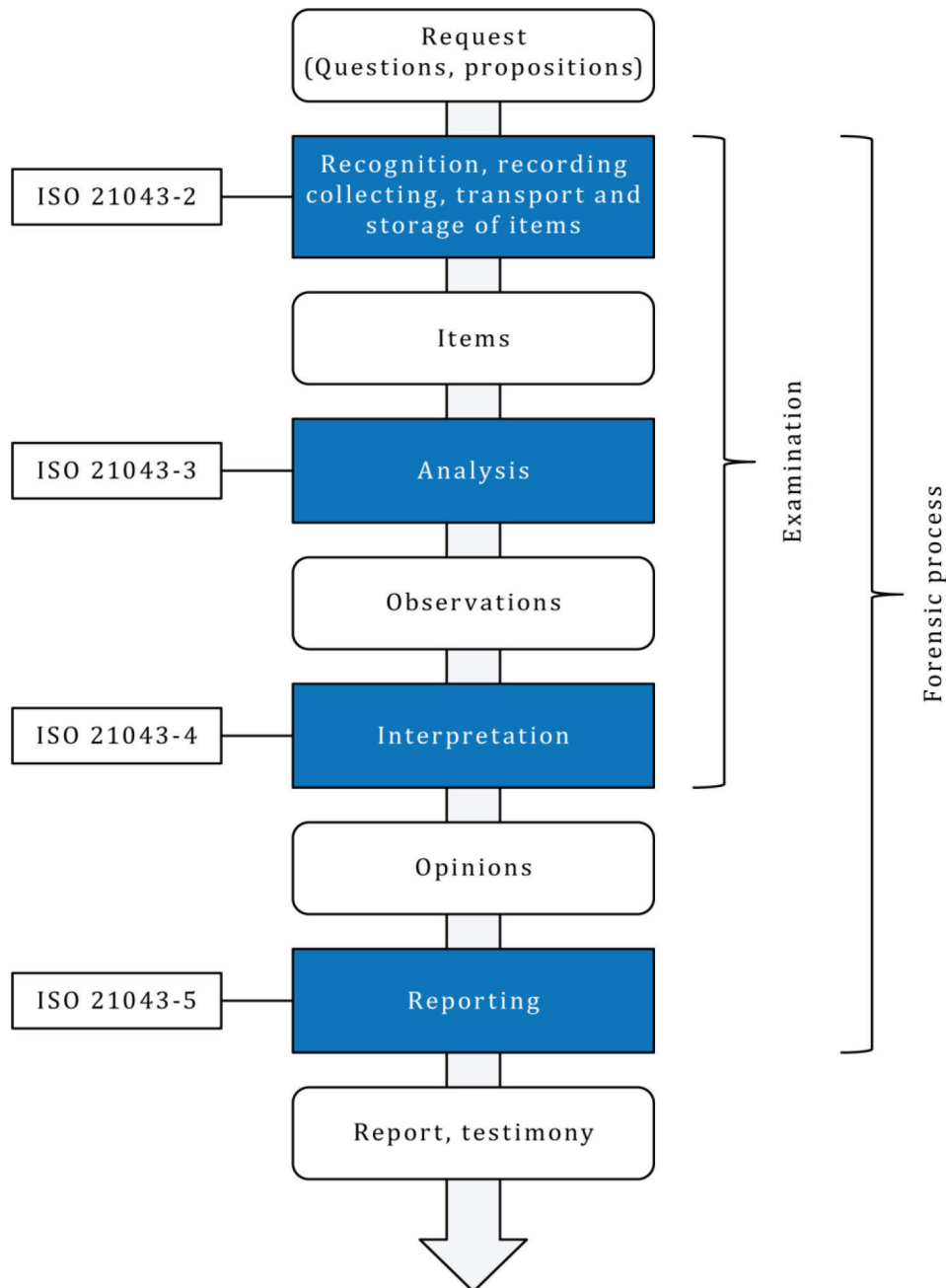


Figure A.1 — Forensic Process

The blue boxes indicate processes and are linked to the various parts of the standard (indicated with labels on the left side). The white boxes with rounded corners can be the input or output of a process.

The forensic process can be very diverse, and the arrow only illustrates the general flow. In a particular forensic process some actions might be skipped or repeated.

Annex B **(informative)**

Reports and other communication of information

This standard distinguishes reports and other communications of information.

Reports are designated as such in the title (e.g., by including the word "report" in the title) and are compliant with the requirements defined in this standard.

Reports can vary in three different, independent aspects:

a) *The content and purpose of the report*

Some reports such as investigative or evaluative reports, contain opinions. Some reports only include observations. Some reports include both observations and opinions. Reports can also be both investigative and evaluative. Reports can include interpretations of observations by providing explanations for the observations or by providing an estimation based on observations. A single report can serve multiple purposes.

b) *The position of the report in an administrative sequence*

Reports can be preliminary, amended, supplemented, withdrawn, or final.

Reports can be a summary of preliminary results or a summary of final results that is provided to customers prior to the production of a final report.

Amended reports are issued when a technical or administrative error is identified, or if a previously issued report needs to be changed. An amended report replaces the original report.

Supplemented reports are issued after additional analysis, or to add information which was not included in the original report.

c) *The channel of delivery of the report*

Reports can be delivered on paper or digitally.

Communications of information other than by way of a report include digital messages, such as e-mails, and oral communication.

E-mails can be used as a channel for issuing a report, but also for digital messages. Digital messages can be used, for example, if the result of a particular test, search or examination was negative, leading to an end of a particular line of enquiry, or a potential candidate was found in a database and further material might be required for analysis.

Oral communication (apart from oral testimony) can be in person, over the phone, or by video conference or other electronic means. Such communications are intended for information and guidance during the course of an investigation or when an examination has yielded negative results leading to the end of an inquiry.